NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | SPCL-GTM-06 | Date: 19.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	B	A	C	C	B	A	A	C	C
11	12	13	14	15	16	17	18	19	20
C	B	D	A	B	A	B	C	B	A
21	22	23	24	25	26	27	28	29	30
2	1	2	5	6					

CHEMISTRY

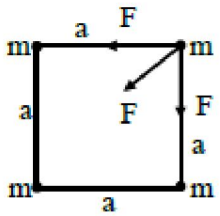
26	27	28	29	30	31	32	33	34	35
B	D	B	A	D	B	C	D	D	B
36	37	38	39	40	41	42	43	44	45
A	A	B	A	A	A	B	C	A	C
46	47	48	49	50					
163	0	4	3	5					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	D	D	B	B	B	B	B	B	D
61	62	63	64	65	66	67	68	69	70
C	D	A	D	D	B	B	D	D	C
71	72	73	74	75					
12	27	16	118	3734					

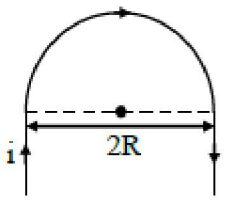
For More Material Join: @JEEAdvanced_2025

PHYSICS

1.	$PV = nRT$ $V = \left(\frac{nR}{P}\right)T$ $\text{Slope} = \frac{nR}{P}$ $\text{Slope} \propto \frac{1}{P}$ $(\text{Slope})_2 > (\text{Slope})_1$ $P_2 < P_1$
2.	 $F_{net} = \sqrt{2}F + F'$ $F = \frac{Gm^2}{a^2}$ and $F' = \frac{Gm^2}{\left(\sqrt{2}a\right)^2}$ $F_{net} = \sqrt{2}\frac{Gm^2}{a^2} + \frac{Gm^2}{2a^2}$ $\left(\frac{2\sqrt{2}+1}{32}\right)\frac{Gm^2}{L^2} = \frac{Gm^2}{a^2}\left(\frac{2\sqrt{2}+1}{2}\right)$ $a = 4L$
3.	$U_E = \frac{1}{2}\epsilon_0 E^2$ $U_E = \frac{1}{2} \times 8.85 \times 10^{-12} \times (50)^2$ $= 1.106 \times 10^{-8}\ \text{J/m}^3$
4.	$\lvert\vec{p_1}\rvert = \lvert\vec{p_2}\rvert$ $KE = \frac{p^2}{2M}$; p same $KE \propto \frac{1}{m}$ $\frac{KE_1}{KE_2} = \frac{p^2/2M_1}{p^2/2M_2} = \frac{M_2}{M_1}$

For More Material Join: @JEEAdvanced_2025

5.



Note : Direction of magnetic field is in $+\hat{k}$

$\vec{F} = i\vec{\ell} \times \vec{B}$

$\ell = 2R$

$\vec{F} = -2iRB\hat{j}$

6.

$$\frac{1}{\lambda} = \frac{13.6z^2}{hc}\left[\frac{1}{1^2} - \frac{1}{2^2}\right] \ldots\ldots \text{(i)}$$

$$\frac{1}{\lambda'} = \frac{13.6z^2}{hc}\left[\frac{1}{1^2} - \frac{1}{3^2}\right] \ldots\ldots \text{(ii)}$$

On dividing (i) & (ii)

$$\lambda' = \frac{27}{32}\lambda$$

7.

$$E = \phi + K_{max}$$

$$\phi = \frac{hc}{\lambda_0}$$

$$K_{max} = eV_0$$

$$8e = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \ldots\ldots \text{(i)}$$

$$2e = \frac{hc}{3\lambda} - \frac{hc}{\lambda_0} \ldots\ldots \text{(ii)}$$

on solving (i) & (ii)

$$\lambda_0 = 9\lambda$$

8.

$$R = \frac{\rho L}{\pi\frac{d^2}{4}}$$

$$\frac{\Delta R}{R} = \frac{\Delta L}{L} + \frac{2\Delta d}{d}$$

$$\frac{\Delta L}{L} = 0.1\% \text{ and } \frac{\Delta d}{d} = 0.1\%$$

$$\frac{\Delta R}{R} = 0.3\%$$

For More Material Join: @JEEAdvanced_2025

9.

$$\varepsilon = N\left(\frac{\Delta\phi}{t}\right)$$

$\Delta\phi = (\Delta B)A$

$B_i = 5000$ T,

$B_f = 3000$ T

$d = 0.02$ m

$r = 0.01$ m

$\Delta\phi = (\Delta B)A$

$= (2000)\pi(0.01)^2 = 0.2\pi$

$$\varepsilon = N\left(\frac{\Delta\phi}{t}\right) \Rightarrow 22 = N\left(\frac{0.2\pi}{2}\right)$$

$N = 70$

10.

$$KE = \frac{f}{2}kT$$

Conceptual

11.

Series :

$R_{eq} = R_1 + R_2$

$2R(1+\alpha_{eq}\Delta\theta) = R(1+\alpha_1\Delta\theta) + R(1+\alpha_2\Delta\theta)$

$2R(1+\alpha_{eq}\Delta\theta) = 2R + (\alpha_1+\alpha_2)R\Delta\theta$

$$\alpha_{eq} = \frac{\alpha_1+\alpha_2}{2}$$

Parallel :

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{\frac{R}{2}(1+\alpha_{eq}\Delta\theta)} = \frac{1}{R(1+\alpha_1\Delta\theta)} + \frac{1}{R(1+\alpha_2\Delta\theta)}$$

$$\frac{2}{1+\alpha_{eq}\Delta\theta} = \frac{1}{1+\alpha_1\Delta\theta} + \frac{1}{1+\alpha_2\Delta\theta}$$

$$\frac{2}{1+\alpha_{eq}\Delta\theta} = \frac{1+\alpha_2\Delta\theta+1+\alpha_1\Delta\theta}{(1+\alpha_1\Delta\theta)(1+\alpha_2\Delta\theta)}$$

$2\left[(1+\alpha_1\Delta\theta)(1+\alpha_2\Delta\theta)\right]$

$= \left[2+(\alpha_1+\alpha_2)\Delta\theta\right]\left[1+\alpha_{eq}\Delta\theta\right]$

$2\left[1+\alpha_1\Delta\theta+\alpha_2\Delta\theta+\alpha_1\alpha_2\Delta\theta\right]$

For More Material Join: @JEEAdvanced_2025

=

$2+2\alpha_{eq}\Delta\theta+(\alpha_1+\alpha_2)\Delta\theta+\alpha_{eq}(\alpha_1+\alpha_2)\Delta\theta^2$

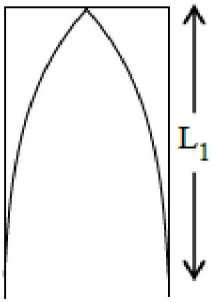
Neglecting small terms

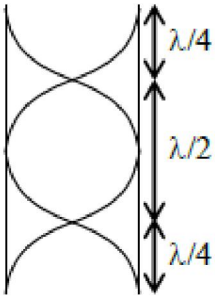
$2+2(\alpha_1+\alpha_2)\Delta\theta=2+2\alpha_{eq}\Delta\theta+(\alpha_1+\alpha_2)\Delta\theta$

$(\alpha_1+\alpha_2)\Delta\theta=2\alpha_{eq}\Delta\theta$

$\alpha_{eq}=\frac{\alpha_1+\alpha_2}{2}$

12.





$\frac{\lambda}{4}=L_1$

$2\left(\frac{\lambda}{2}\right)=\lambda$

$v=f\lambda$

$f_2=\frac{2v}{2L_2}$

$v=f_1(4L_1)$

$f_2=\frac{v}{L_2}$

$f_1=\frac{v}{4L_1}$

$f_1=f_2$

$\frac{v}{4L_1}=\frac{v}{L_2}$

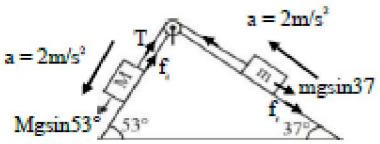
$\Rightarrow L_2=4L_1$

$60=4\times L_1$

$L_1=15$ cm

For More Material Join: @JEEAdvanced_2025

13.



For M block

$10g\sin53° - \mu(10g)\cos53° - T = 10 \times 2$

$T = 80 - 15 - 20$

$T = 45\ N$

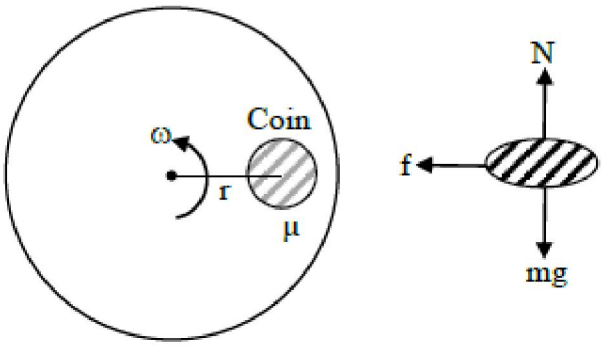
For m block

$T - mg\sin37° - \mu mg\cos37° = m \times 2$

$45 = 10\,m$

$m = 4.5\ kg$

14.



$N = mg$

$f = m\omega^2 r$

$f = \mu N$

$\mu mg = mr\omega^2$

$$\omega = \sqrt{\frac{\mu g}{r}}$$

15.

$F = ax^2 + bt^{1/2}$

$$[a] = \frac{[F]}{[x^2]} = [M^1L^{-1}T^{-2}]$$

$$[b] = \frac{[F]}{[t^{1/2}]} = [M^1L^1T^{-5/2}]$$

$$\left[\frac{b^2}{a}\right] = \frac{[M^2L^2T^{-5}]}{[M^1L^{-1}T^{-2}]} = [M^1L^3T^{-3}]$$

For More Material Join: @JEEAdvanced_2025

16.

$t = \alpha x^2 + \beta x$ (differentiating wrt time)

$$\frac{dt}{dx} = 2\alpha x + \beta$$

$$\frac{1}{v} = 2\alpha x + \beta$$

(differentiating wrt time)

$$-\frac{1}{v^2}\frac{dv}{dt} = 2\alpha \frac{dx}{dt}$$

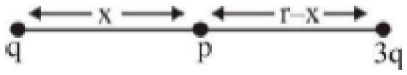
$$\frac{dv}{dt} = -2\alpha v^3$$

17.

$Y = \overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A}} + \overline{\overline{B}} = A + B$

(De-Morgan's law)

18.



$$\left(\vec{E}_{net}\right)_p = 0$$

$$\frac{kq}{x^2} = \frac{k \cdot 3q}{(r-x)^2}$$

$$(r-x)^2 = 3x^2$$

$$r - x = \sqrt{3}x$$

$$x = \frac{r}{\sqrt{3}+1}$$

19.

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\frac{A}{2}}$$

$$\frac{\cos\frac{A}{2}}{\sin\frac{A}{2}} = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\frac{A}{2}}$$

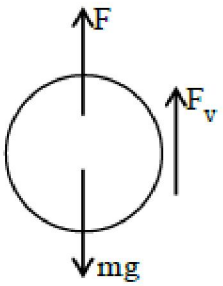
$$\sin\left(\frac{\pi}{2} - \frac{A}{2}\right) = \sin\left(\frac{A+\delta_m}{2}\right)$$

$$\frac{\pi}{2} - \frac{A}{2} = \frac{A}{2} + \frac{\delta m}{2}$$

$$\delta_m = \pi - 2A$$

For More Material Join: @JEEAdvanced_2025

20.



$$mg - F_B - F_v = ma$$

$$\left(\rho\frac{4}{3}\pi r^3\right)g - \left(\rho_L\frac{4}{3}\pi r^3\right)g - 6\pi\eta rv = m\frac{dv}{dt}$$

Let $\frac{4}{3m}\pi R^3 g(\rho-\rho_L) = K_1$ and $\frac{6\pi\eta r}{m} = K_2$

$$\frac{dv}{dt} = K_1 - K_2 v$$

$$\int_0^v \frac{dv}{K_1 - K_2 v} = \int_0^t dt$$

$$-\frac{1}{K_2}\ell n\left[K_1 - K_2 v\right]_0^v = t$$

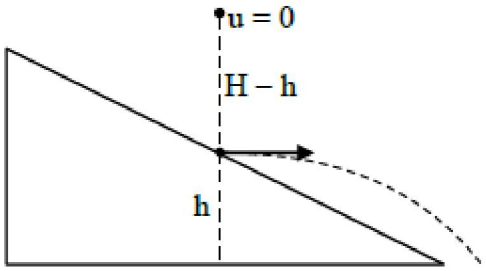
$$\ell n\left(\frac{K_1 - K_2 v}{K_1}\right) = -K_2 t$$

$$K_1 - K_2 v = K_1 e^{-K_2 t}$$

$$v = \frac{K_1}{K_2}\left[1 - e^{-K_2 t}\right]$$

For More Material Join: @JEEAdvanced_2025

21.



Total time of flight = T

$$T = \sqrt{\frac{2h}{g}} + \sqrt{\frac{2(H-h)}{g}}$$

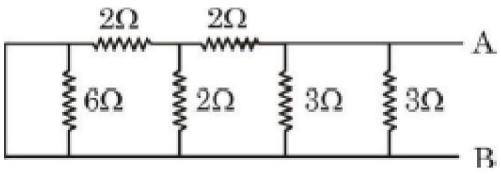
For max. time $= \frac{dT}{dh} = 0$

$$\sqrt{\frac{2}{g}}\left(\frac{-1}{2\sqrt{H-h}} + \frac{1}{2\sqrt{h}}\right) = 0$$

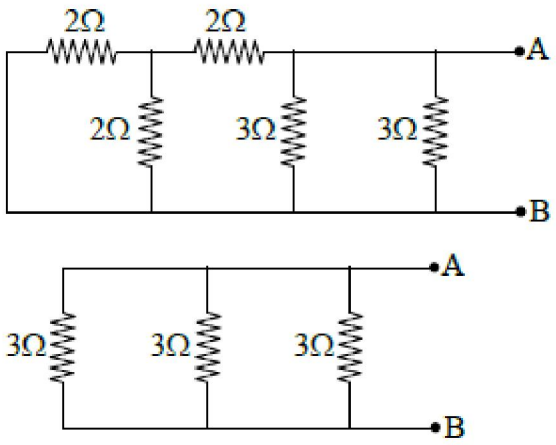
$$\sqrt{H-h} = \sqrt{h}$$

$$h = \frac{H}{2} \Rightarrow \frac{H}{h} = 2$$

22.



6Ω is short circuit



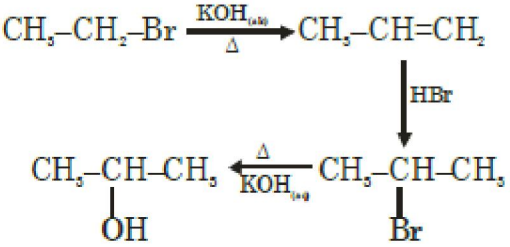
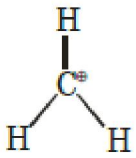
$$R_{eq} = 3 \times \frac{1}{3} = 1\Omega$$

For More Material Join: @JEEAdvanced_2025

23.	Without dielectric $Q = \frac{A\epsilon_0}{d}V$ with dielectric $Q = \frac{A\epsilon_0 V}{d - t + \frac{t}{K}}$ given $\frac{A\epsilon_0 V}{d - t + \frac{t}{K}} = (1.25)\frac{A\epsilon_0 V}{d}$ $\Rightarrow 1.25\left(3 + \frac{2}{K}\right) = 5$ $\Rightarrow K = 2$
24.	$\vec{F} = q(\vec{v} \times \vec{B})$ $5e\hat{k} = e(3\hat{i} + 5\hat{j}) \times (B_0\hat{i} + 2B_0\hat{j})$ $5e\hat{k} = e(6B_0\hat{k} - 5B_0\hat{k})$ $\Rightarrow B_0 = 5T$
25.	Using work energy theorem $W = \Delta KE = 0 - \left(\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2\right)$ $W = 0 - \frac{1}{2}mv^2\left(1 + \frac{K^2}{R^2}\right)$ $= -\frac{1}{2} \times 50 \times 0.4^2\left(1 + \frac{1}{2}\right) = -6J$ Absolute work = +6J $W = -6J \quad |W| = 6J$

For More Material Join: @JEEAdvanced_2025

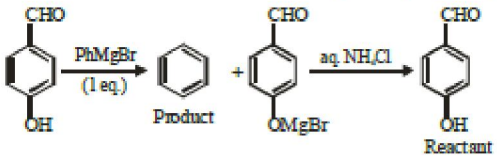
CHEMISTRY

26.	Element $\Delta_{eg}H$(kJ/mol) F –333 S –200 Br –325 Ar +96
27.	CaF_2 does not evolve any gas with concentrated H_2SO_4. NaBr → evolve Br_2 $NaNO_3$ → evolve NO_2 KI → evolve I_2
28.	SnO_2 and PbO_2 are amphoteric
29.	Principle used in chromotography is adsorption
30.	7-Hydroxyheptan-2-one is correct IUPAC name
31.	
32.	As per NCERT, Assertion (A) and Reason (R) is correct but Reason (R) is not the correct explanation.
33.	 Six electron species
34.	Lead sulphate-white Ammonium sulphide-soluble Lead iodide-Bright yellow Ammonium arsinomolybdate-yellow
35.	$(CH_3)_2CO + CS_2$ Exibits positive deviations from Raoult's Law

For More Material Join: @JEEAdvanced_2025

36.	Conductivity of electrolytic cell is affected by concentration of electrolyte, nature of electrolyte and nature of solvent.
37.	$E = \frac{1240}{\lambda(nm)} eV$ $= \frac{1240}{242} eV$ $= 5.12\ eV$ $= 5.12 \times 1.6 \times 10^{-19}$ $= 8.198 \times 10^{-19}$ J/atom $= 494$ kJ/mol
38.	Mn, Ni and Cd metals used in battery industries.
39.	$K_C = \frac{\text{Products ion conc.}}{\text{Reactants ion conc.}}$ $K_C = \frac{[FeSCN^{2+}]}{[Fe^{3+}][SCN^-]}$
40.	* Molecular orbital should have maximum overlap * Symmetry about the molecular axis should be similar
41	Statement I and II are False Noble gases have low boiling points Noble gases are held together by weak dispersion forces.
42.	$A \rightarrow B + C$ $P_i \quad 0 \quad 0$ $P_i - x \quad x \quad x$ $P_t = P_i + x$ $P_i - x = P_i - P_t + P_i$ $= 2P_i - P_t$ $K = \frac{2.303}{t} \log \frac{P_i}{2P_i - P_t}$
43.	A. CrO_4^{2-} is tetrahedral B. $2Na_2CrO_4 + 2H^+ \rightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$ C. As per NCERT, green manganate is paramagnetic with 1 unpaired electron. D. Statement is correct E. Statement is correct

For More Material Join: @JEEAdvanced_2025

44.	Phenol is more acidic than ethanol because conjugate base of phenoxide is more stable than ethoxide.
45.	Glucose $\xrightarrow[\Delta]{NaHCO_3}$ no reaction Glucose $\xrightarrow[\Delta]{HNO_3}$ saccharic acid Glucose $\xrightarrow[\Delta]{HI}$ n-hexane Glucose $\xrightarrow[\Delta]{Br_2}$ Gluconic acid
46.	$\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)}.K_p = 2.47\times10^{-29}.$ $\Delta_r G^{\ominus} = -RT \ln K_P$ $= -8.314 \times 10^{-3} \times 298 \times \ln (2.47 \times 10^{-29})$ $= -8.314 \times 10^{-3} \times 298 \times (-65.87)$ $= 163.19$ kJ
47.	Product benzene has zero hydroxyl group 
48.	$NH_3 \rightarrow sp^3$ $SO_2 \rightarrow sp^2$ $SiO_2 \rightarrow sp^3$ $BeCl_2 \rightarrow sp$ $CO_2 \rightarrow sp$ $H_2O \rightarrow sp^3$ $CH_4 \rightarrow sp^3$ $BF_3 \rightarrow sp^2$
49.	$CH_3COONa \rightarrow \dot{C}H_3$ $C_2H_5COONa \rightarrow \dot{C}_2H_5$ $2\dot{C}_2H_5 \rightarrow CH_3-CH_2-CH_2-CH_3$ $2\dot{C}H_3 \rightarrow CH_3-CH_3$ $\dot{C}H_3 + \dot{C}_2H_5 \rightarrow CH_3-CH_2-CH_3$

For More Material Join: @JEEAdvanced_2025

50.	$Cu^{2+} + 2e^- \rightarrow Cu$ 2 Faraday $\rightarrow$ 1 mol Cu 1 Faraday $\rightarrow$ 0.5 mol Cu deposit 0.5 mol = 0.5 g atom = 5×10^{-1} x = 5

For More Material Join: @JEEAdvanced_2025

MATHS

51.

$$f(0)=\begin{vmatrix}0 & 1 & 1\\ 2 & 0 & 6\\ 0 & 4 & -2\end{vmatrix}=12$$

$$f'(x)=\begin{vmatrix}3x^2 & 4x & 3\\ 3x^2+2 & 2x & x^3+6\\ x^3-x & 4 & x^2-2\end{vmatrix}+$$

$$\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 6x & 2 & 3x^2\\ x^3-x & 4 & x^2-2\end{vmatrix}+$$

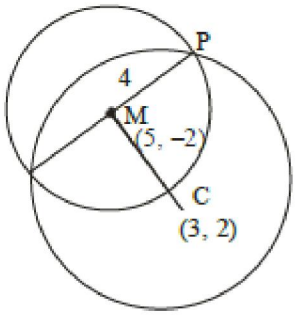
$$\begin{vmatrix}x^3 & 2x^2+1 & 1+3x\\ 3x^2+2 & 2x & x^3+6\\ 3x^2-1 & 0 & 2x\end{vmatrix}$$

$$\therefore f'(0)=\begin{vmatrix}0 & 0 & 3\\ 2 & 0 & 6\\ 0 & 4 & -2\end{vmatrix}+\begin{vmatrix}0 & 1 & 1\\ 0 & 2 & 0\\ 0 & 4 & -2\end{vmatrix}+\begin{vmatrix}0 & 1 & 1\\ 2 & 0 & 6\\ -1 & 0 & 0\end{vmatrix}$$

$=24-6=18$

$\therefore 2f(0)+f'(0)=42$

52.



$2x+3y=12$

$3x-2y=5$

$13x=39$

$x=3, y=2$

Center of given circle is $(5,-2)$

Radius $\sqrt{25+4-13}=4$

$\therefore CM=\sqrt{4+16}=5\sqrt{2}$

$\therefore CP=\sqrt{16+20}=6$

For More Material Join: @JEEAdvanced_2025

53.

$$\frac{dy}{dx}=\frac{\sin x+y\cos x}{\sin x.\cos x\left(\frac{1}{\cos x}-\sin x.\frac{\sin x}{\cos x}\right)}$$

$$=\frac{\sin x+y\cos x}{\sin x\left(1-\sin^2 x\right)}$$

$$\frac{dy}{dx}=\sec^2 x+y.2\left(\operatorname{cosec}2x\right)$$

$$\frac{dy}{dx}-2\operatorname{cosec}\left(2x\right).y=\sec^2 x$$

$$\frac{dy}{dx}+p.y=Q$$

$$\text{I.F.}=e^{\int p dx}=e^{\int -2\operatorname{cosec}(2x)dx}$$

Let $2x=t$

$$2\frac{dx}{dt}=1$$

$$dx=\frac{dt}{2}$$

$$=e^{-\int \operatorname{cosec}(t)dt}$$

$$=e^{-\ln\left|\tan\frac{t}{2}\right|}$$

$$=e^{-\ln|\tan x|}=\frac{1}{|\tan x|}$$

$$y(\text{IF})=\int Q(\text{IF})dx+c$$

$$\Rightarrow y\frac{1}{|\tan x|}=\int \sec^2 x\cdot\frac{1}{|\tan x|}+c$$

$$y.\frac{1}{|\tan x|}=\int\frac{dt}{|t|}+c \qquad \text{for } \tan x=t$$

$$y.\frac{1}{|\tan x|}=\ln|t|+c$$

$$y=|\tan x|\left(\ln|\tan x|+c\right)$$

Put $x=\frac{\pi}{4},\ y=2$

$$2=\ln 1+c \Rightarrow c=2$$

$$y=|\tan x|\left(\ln|\tan x|+2\right)$$

$$y\left(\frac{\pi}{3}\right)=\sqrt{3}\left(\ln\sqrt{3}+2\right)$$

For More Material Join: @JEEAdvanced_2025

54.

$f(x) = (a + b - 2c)x^2 + (b + c - 2a)x + (c + a - 2b)$

$f(x) = a + b - 2c + b + c - 2a + c + a - 2b = 0$

$f(1) = 0$

$$\therefore \alpha \cdot 1 = \frac{c + a - 2b}{a + b - 2c}$$

$$\alpha = \frac{c + a - 2b}{a + b - 2c}$$

If, $-1 < \alpha < 0$

$$-1 < \frac{c + a - 2b}{a + b - 2c} < 0$$

$$b + c < 2a \text{ and } b > \frac{a + c}{2}$$

therefore, b cannot be G.M. between a and c.

If, $0 < \alpha < 1$

$$0 < \frac{c + a - 2b}{a + b - 2c} < 1$$

$$b > c \text{ and } b < \frac{a + c}{2}$$

Therefore, b may be the G.M. between a and c.

55.

Let $g(x) = ax + b$

Now function f(x) in continuous at x = 0

$$\therefore \lim_{x \to 0^+} f(x) = f(0)$$

$$\Rightarrow \lim_{x \to 0} \left(\frac{1 + x}{2 + x} \right)^{\frac{1}{x}} = b$$

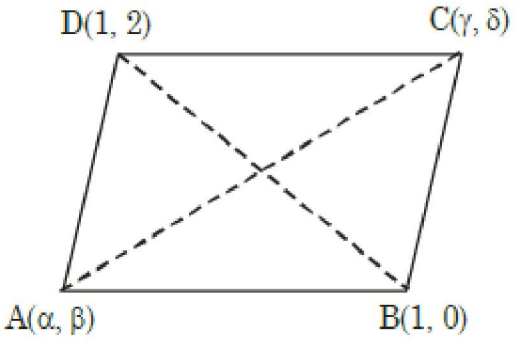
$$\Rightarrow 0 = b$$

$$\therefore g(x) = ax$$

Now, for $x > 0$

$$f'(x) = \frac{1}{x} \cdot \left(\frac{1 + x}{2 + x} \right)^{\frac{1}{x} - 1} \cdot \frac{1}{(2 + x)^2}$$

For More Material Join: @JEEAdvanced_2025

	$+\left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}\cdot\ln\left(\frac{1+x}{2+x}\right)\cdot\left(-\frac{1}{x^2}\right)$ $\therefore f'(1)=\frac{1}{9}-\frac{2}{3}\cdot\ln\left(\frac{2}{3}\right)$ And $f(-1)=g(-1)=-a$ $\therefore a=\frac{2}{3}\ln\left(\frac{2}{3}\right)-\frac{1}{9}$ $\therefore g(3)=2\ln\left(\frac{2}{3}\right)-\frac{1}{3}$ $=\ln\left(\frac{4}{9\cdot e^{1/3}}\right)$
56.	 Let E is mid point of diagonals $\frac{\alpha+\gamma}{2}=\frac{1+1}{2}$ & $\frac{\beta+\delta}{2}=\frac{2+0}{2}$ $\alpha+\gamma=2$ $\beta+\delta=2$ $2(\alpha+\beta+\gamma+\delta)=2(2+2)=8$
57.	Probability of drawing first red and then white $=\frac{10}{75}\times\frac{30}{75}=\frac{4}{75}$
58.	$\lim_{x\to0}\frac{e^{2\lvert\sin x\rvert}-2\lvert\sin x\rvert-1}{x^2}$ $\lim_{x\to0}\frac{e^{2\lvert\sin x\rvert}-2\lvert\sin x\rvert-1}{\lvert\sin x\rvert^2}\times\frac{\sin^2 x}{x^2}$ Let $\lvert\sin x\rvert=t$ $\lim_{t\to0}\frac{e^{2t}-2t-1}{t^2}\times\lim_{x\to0}\frac{\sin^2 x}{x^2}$ $=\lim_{t\to0}\frac{2e^{2t}-2}{2t}\times1=2\times1=2$

For More Material Join: @JEEAdvanced_2025

59.

$$\frac{dx}{dy}=\frac{x}{y}\left(\ln\left(\frac{x}{y}\right)+1\right)$$

Let $\frac{x}{y}=t\Rightarrow x=ty$

$$\frac{dx}{dy}=t+y\frac{dt}{dy}$$

$$t+y\frac{dt}{dy}=t\left(\ln(t)+1\right)$$

$$y\frac{dt}{dy}=t\ln(t)\Rightarrow\frac{dt}{t\ln(t)}=\frac{dy}{y}$$

$$\Rightarrow\int\frac{dt}{t.\ln(t)}=\int\frac{dy}{y}$$

$$\Rightarrow\int\frac{dp}{p}=\int\frac{dy}{y}\qquad \text{let } \ln t=p$$

$$\frac{1}{t}dt=dp$$

$$\Rightarrow \ln p=\ln y+c$$

$$\ln(\ln t)=\ln y+c$$

$$\ln\left(\ln\left(\frac{x}{y}\right)\right)=\ln y+c$$

at $x=e, y=1$

$$\ln\left(\ln\left(\frac{e}{1}\right)\right)=\ln(1)+c\Rightarrow c=0$$

$$\ln\left|\ln\left(\frac{x}{y}\right)\right|=\ln y$$

$$\left|\ln\left(\frac{x}{y}\right)\right|=e^{\ln y}$$

$$\left|\ln\left(\frac{x}{y}\right)\right|=y$$

For More Material Join: @JEEAdvanced_2025

60.

Put $x = 1$

$\therefore a = 1$

$$b = \lim_{x\to 0}\frac{\int_0^x \frac{\ln(1+t)}{1+t^{2024}}dt}{x^2}$$

Using L' HOPITAL Rule

$$b = \lim_{x\to 0}\frac{\ln(1+x)}{(1+x^{2024})}\times\frac{1}{2x} = \frac{1}{2}$$

Now, $cx^2 + dx + e = 0$, $x^2 + x + 4 = 0$

$(D < 0)$

$$\therefore \frac{c}{1} = \frac{d}{1} = \frac{e}{4}$$

61.

A vector in the direction of the required line can be obtained by cross product of

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 5 \\ -1 & 3 & 2 \end{vmatrix}$$

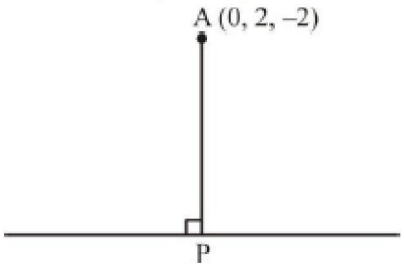
$$= -9\hat{i} - 9\hat{j} + 9\hat{k}$$

Required line,

$$\vec{r} = (5\hat{i} - 4\hat{j} + 3\hat{k}) + \lambda'(-9\hat{i} - 9\hat{j} + 9\hat{k})$$

$$\vec{r} = (5\hat{i} - 4\hat{j} + 3\hat{k}) + \lambda(\hat{i} + \hat{j} - \hat{k})$$

Now distance of (0, 2, –2)



P.V. of $P \equiv (5+\lambda)\hat{i} + (\lambda - 4)\hat{j} + (3-\lambda)\hat{k}$

$$\overrightarrow{AP} = (5+\lambda)\hat{i} + (\lambda - 6)\hat{j} + (5-\lambda)\hat{k}$$

$$\overrightarrow{AP}\cdot(\hat{i} + \hat{j} - \hat{k}) = 0$$

$$5 + \lambda + \lambda - 6 - 5 + \lambda = 0$$

$$\lambda = 2$$

$$\left|\overrightarrow{AP}\right| = \sqrt{49 + 16 + 9}$$

$$\left|\overrightarrow{AP}\right| = \sqrt{74}$$

For More Material Join: @JEEAdvanced_2025

62.

General term of the sequence,

$$T_r = \frac{r}{1-3r^2+r^4}$$

$$T_r = \frac{r}{r^4-2r^2+1-r^2}$$

$$T_r = \frac{r}{(r^2-1)^2-r^2}$$

$$T_r = \frac{r}{(r^2-r-1)(r^2+r-1)}$$

$$T_r = \frac{\frac{1}{2}\left[(r^2+r-1)-(r^2-r-1)\right]}{(r^2-r-1)(r^2+r-1)}$$

$$= \frac{1}{2}\left[\frac{1}{r^2-r-1}-\frac{1}{r^2+r-1}\right]$$

Sum of 10 terms,

$$\sum_{r=1}^{10} T_r = \frac{1}{2}\left[\frac{1}{-1}-\frac{1}{109}\right] = \frac{-55}{109}$$

63.

Let $\sin^{-1}\alpha = A, \sin^{-1}\beta = B, \sin^{-1}\gamma = C$

$A + B + C = \pi$

$(\alpha+\beta)^2 - \gamma^2 = 3\alpha\beta$

$\alpha^2 + \beta^2 - \gamma^2 = \alpha\beta$

$$\frac{\alpha^2+\beta^2-\gamma^2}{2\alpha\beta} = \frac{1}{2}$$

$$\Rightarrow \cos C = \frac{1}{2}$$

$\sin C = \gamma$

$$\cos C = \sqrt{1-\gamma^2} = \frac{1}{2}$$

$$\gamma = \frac{\sqrt{3}}{2}$$

For More Material Join: @JEEAdvanced_2025

64.

$\vec{p}\times\vec{b}-\vec{c}\times\vec{b}=\vec{0}$

$(\vec{p}-\vec{c})\times\vec{b}=\vec{0}$

$\vec{p}-\vec{c}=\lambda\vec{b}\Rightarrow\vec{p}=\vec{c}+\lambda\vec{b}$

Now, $\vec{p}.\vec{a}=0(\text{given})$

So, $\vec{c}.\vec{a}+\lambda\vec{a}.\vec{b}=0$

$(3-3-8)+\lambda(12+1-14)=0$

$\lambda=-8$

$\vec{p}=\vec{c}-8\vec{b}$

$\vec{p}=-31\hat{i}-11\hat{j}-52\hat{k}$

So, $\vec{p}.(\hat{i}-\hat{j}-\hat{k})$

$=-31+11+52$

$=32$

65.

$$D=\begin{vmatrix}1 & -2 & 1\\ 2 & \alpha & 3\\ 3 & -1 & \beta\end{vmatrix}$$

$=1(\alpha\beta+3)+2(2\beta-9)+1(-2-3\alpha)$

$=\alpha\beta+3+4\beta-18-2-3\alpha$

For infinite solutions $D=0$, $D_1=0$, $D_2=0$ and $D_3=0$

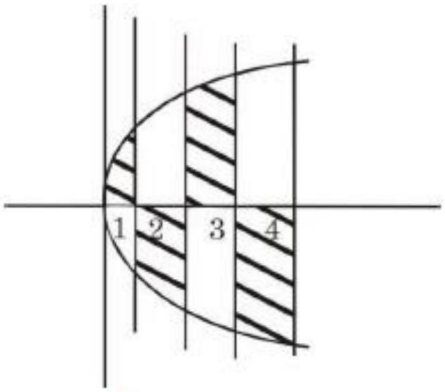
$D=0$

$\alpha\beta-3\alpha+4\beta=17$ ....(1)

$$D_1=\begin{vmatrix}-4 & -2 & 1\\ 5 & \alpha & 3\\ 3 & -1 & \beta\end{vmatrix}=0$$

$$D_2=\begin{vmatrix}1 & -4 & 1\\ 2 & 5 & 3\\ 3 & 3 & \beta\end{vmatrix}=0$$

For More Material Join: @JEEAdvanced_2025

	$\Rightarrow 1(5\beta-9)+4(2\beta-9)+1(6-15)=0$ $13\beta-9-36-9=0$ $13\beta=54, \beta=\frac{54}{13}$ put in (1) $\frac{54}{13}\alpha-3\alpha+4\left(\frac{54}{13}\right)=17$ $54\alpha-39\alpha+216=221$ $15\alpha=5 \quad \alpha=\frac{1}{3}$ Now, $12\alpha+13\beta=12.\frac{1}{3}+13.\frac{54}{13}$ $= 4+54=58$
66.	$y^2 \le 4x, x<4$ $\frac{xy(x-1)(x-2)}{(x-3)(x-4)}>0$ Case – I : $y>0$ $\frac{x(x-1)(x-2)}{(x-3)(x-4)}>0$ $x\in(0,1)\cup(2,3)$ Case – II : $y<0$ $\frac{x(x-1)(x-2)}{(x-3)(x-4)}<0, x\in(1,2)\cup(3,4)$  Area $=2\int_0^4 \sqrt{x}\,dx$ $=2\cdot\frac{2}{3}\left[x^{3/2}\right]_0^4=\frac{32}{3}$

For More Material Join: @JEEAdvanced_2025

67.	3 bad apples, 15 good apples. Let X be no of bad apples Then $P(X=0)=\frac{{}^{15}C_2}{{}^{18}C_2}=\frac{105}{153}$ $P(X=1)=\frac{{}^{3}C_1\times{}^{15}C_1}{{}^{18}C_2}=\frac{45}{153}$ $P(X=2)=\frac{{}^{3}C_2}{{}^{18}C_2}=\frac{3}{153}$ $E(X)=0\times\frac{105}{153}+1\times\frac{45}{153}+2\times\frac{3}{153}=\frac{51}{153}$ $=\frac{1}{3}$ $Var(X)=E(X^2)-(E(X))^2$ $=0\times\frac{105}{153}+1\times\frac{45}{153}+4\times\frac{3}{153}-\left(\frac{1}{3}\right)^2$ $=\frac{57}{153}-\frac{1}{9}=\frac{40}{153}$
68.	$f(x)=\frac{4x+3}{6x-4}$ $g(x)=\frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}=\frac{34x}{34}=x$ $g(x)=x\therefore g(g(g(4)))=4$

For More Material Join: @JEEAdvanced_2025

69.

$\frac{x^2}{9}+\frac{y^2}{25}=1$

$a=3, b=5$

$e=\sqrt{1-\frac{9}{25}}=\frac{4}{5} \quad \therefore \text{foci}=(0,\pm be)=(0,\pm 4)$

$\therefore e_H=\frac{4}{5}\times\frac{15}{8}=\frac{3}{2}$

Let equation hyperbola

$\frac{x^2}{A^2}-\frac{y^2}{B^2}=-1$

$\therefore B\cdot e_H=4 \quad \therefore B=\frac{8}{3}$

$\therefore A^2=B^2\left(e_H^2-1\right)=\frac{64}{9}\left(\frac{9}{4}-1\right) \quad \therefore A^2=\frac{80}{9}$

$\therefore \frac{x^2}{\frac{80}{9}}-\frac{y^2}{\frac{64}{9}}=-1$

Directrix : $y=\pm\frac{B}{e_H}=\pm\frac{16}{9}$

$PS=e\cdot PM=\frac{3}{2}\left|\frac{14}{3}\cdot\sqrt{\frac{2}{5}}-\frac{16}{9}\right|$

$=7\sqrt{\frac{2}{5}}-\frac{8}{3}$

70.

$ax^2+2(a+1)x+9a+4<0 \quad \forall x\in R$

$\therefore a<0$

71.

$\frac{x}{1}=\frac{y}{1}=\frac{z-1}{0}=r\rightarrow Q(r,r,1)$

$\frac{x}{1}=\frac{y}{-1}=\frac{z+1}{0}=k\rightarrow R(k,-k,-1)$

$\overrightarrow{PQ}=(a-r)\hat{i}+(a-r)\hat{j}+(a-1)\hat{k}$

$a=r+a-r=0.$

$2a=2r\rightarrow a=r$

$\overrightarrow{PR}=(a-k)i+(a+k)\hat{j}+(a+1)\hat{k}$

$a-k-a-k=0 \Rightarrow k=0$

As, $PQ\perp PR$

$(a-r)(a-k)+(a-r)(a+k)+(a-1)(a+1)=0$

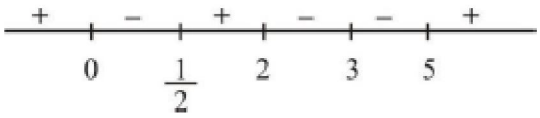
$a=1 \text{ or } -1$

$12a^2=12$

For More Material Join: @JEEAdvanced_2025

72.

$f'(x)=(e^x-1)^{11}(2x-1)^5(x-2)^7(x-3)^{12}(2x-10)^{61}$



Local minima at $x=\frac{1}{2}$, $x=5$

Local maxima at $x=0$, $x=2$

$\therefore p=0+4=4, q=\frac{1}{2}+5=\frac{11}{2}$

Then $p^2+2q=16+11=27$

73.

All elements are included

Answer is 16

74.

$(1+x)(1-x^2)\left(1+\frac{3}{x}+\frac{3}{x^2}+\frac{1}{x^3}\right)^5$

$=(1+x)(1-x^2)\left(\left(1+\frac{1}{x}\right)^3\right)^5$

$=\frac{(1+x)^2(1-x)(1+x)^{15}}{x^{15}}$

$=\frac{(1+x)^{17}-x(1+x)^{17}}{x^{15}}$

$=\text{coeff}\left(x^3\right)$ in the expansion $\approx \text{coeff}\left(x^{18}\right)$ in $(1+x)^{17}-x(1+x)^{17}$

$=0-1$

$=-1$

$\text{coeff}\left(x^{-13}\right)$ in the expansion $\approx \text{coeff}\left(x^2\right)$ in $(1+x)^{17}-x(1+x)^{17}$

$=\binom{17}{2}-\binom{17}{1}$

$=17\times 8-17$

$=17\times 7$

$=119$

Hence Answer = 119 – 1 = 118

For More Material Join: @JEEAdvanced_2025

75.	We have III, TT, D, S, R, B, U, O, N Number of words with selection (a, a, a, b) $= {}^{8}C_1 \times \frac{4!}{3!} = 32$ Number of words with selection (a, a, b, b) $= \frac{4!}{2!2!} = 6$ Number of words with selection (a, a, b, c) $= {}^{2}C_1 \times {}^{8}C_2 \times \frac{4!}{2!} = 672$ Number of words with selection (a, b, c, d) $= {}^{9}C_4 \times 4! = 3024$ $\therefore$ total = 3024 + 672 + 6 + 32 = 3734

For More Material Join: @JEEAdvanced_2025